

Run 03 | NME

Global (including India)

IV only

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 69,931 (5.21%) **Case time** 52m 58s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9763). Highest F1: Random Forest (0.6332). The all-negative accuracy baseline is 94.79% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9763	0.6786	95.87	59.14	67.22	0.6293	20s
CatBoost	0.9756	0.6711	95.80	58.52	66.61	0.6230	1m 35s
Random Forest	0.9755	0.6794	95.91	59.37	67.82	0.6332	2m 38s
H2O GBM	0.9745	0.6588	95.68	57.45	65.76	0.6132	1m 21s
Elastic Net	0.9680	0.5581	94.30	46.85	71.04	0.5647	13m 29s
Lasso	0.9680	0.5579	94.30	46.87	70.92	0.5644	17m 20s
Logistic Regression	0.9677	0.5523	93.85	44.72	76.88	0.5655	2m 26s
Decision Tree	0.9657	0.5778	95.82	62.88	48.11	0.5451	11s
Ridge	0.9655	0.5154	93.73	44.05	75.90	0.5575	9m 07s
Naive Bayes	0.8742	0.2087	88.68	24.11	54.64	0.3345	4m 03s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	17,902.80	37.59
2	Established business owner	6,609.64	13.88
3	Market access and infrastructure	1,394.38	2.93
4	Education	1,377.84	2.89
5	Living conditions	1,274.45	2.68
6	Self-efficacy	1,119.66	2.35
7	Safety and security	1,102.75	2.32
8	Prosperity index score	1,098.90	2.31

Share is the percentage of total forest impurity importance across all 51 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Market access and infrastructure	1,394.38	2.93
2	Education	1,377.84	2.89
3	Living conditions	1,274.45	2.68

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 03_obj2_global_iv_NME. Scores and importance are taken directly from saved CSV outputs.